AMAN – Sprint 0 POC Results (Android)

POC 1: Values-based Blur · POC 2: Core Protection and Anti-Bypass Engine

Result: POC 1 – **Go with limitation** | POC 2 – **Go**

Tested on a Google Pixel 4 with representative scenarios. Testing on other brands (Samsung, Xiaomi and others) belongs to the full development phase. **"Alert only"** means we detect it and alert the parent, but Android does not let us stop it.

POC 2 – Core Protection and Anti-Bypass Engine

Area	What the app does	Limitation
Harmful website blocking	Blocks harmful sites in every browser and app, on Wi-Fi and mobile data, using a filter that runs on the phone (local VPN). Ready-made categories (Adult, Gambling, Violence, Scams, Dating, Social, Streaming). Closes searches for restricted words typed into a browser.	Blocks whole websites, not single pages. A site opened by its number address (IP) or through a web proxy is not blocked. The built-in site list is a demo list, so production needs a licensed list with Arabic coverage.
Safe Search	Forced on for Google (all country sites), Bing, DuckDuckGo and YouTube (app and website), in every browser including incognito. The child cannot switch it off.	Works only while our filter is running. Smaller search engines are not covered.
Block-list / Allow-list	The parent adds or removes sites, with no size limit. An "allow-list only" mode opens only the listed sites while updates, notifications and sign-in keep working. Lists are saved on the phone.	Some allowed sites may be missing pictures until the server those pictures come from is also allowed. Clearing app data erases the lists (a backend would fix this).
VPN detection / restriction	Finds VPN apps and closes them when they open. Any other VPN that connects is detected within seconds and switched off, because Android allows only one VPN and we take the slot back. The parent is alerted. Banking and government ID apps (Absher, UAE Pass and others) are kept outside the filter so they keep working.	**Alert only:** another VPN set to "Always-on" in Settings, or the child removing our VPN permission.
DNS / network bypass	Changing the phone's DNS, or turning on Chrome's "Secure DNS", does not bypass blocking. Android's Private DNS setting is checked every 2 seconds. The parent can choose VPN filter, Private DNS or Hybrid mode (Hybrid is recommended).	**Alert only:** the child changing Private DNS in Settings. **Not blocked:** little-known private DNS servers, and devices connected to the child's hotspot.
Internet Pause / Resume	One tap pauses or resumes the internet on the whole phone (Wi-Fi and mobile data). The child sees a clear "paused by parent" notice. The pause stays in place through restarts and app kills. Calls and SMS still work.	Needs the VPN permission. **Alert only:** the child removing that permission ends the pause.
Restart and tampering	Protection restarts after a reboot. The dashboard shows health (Active / Degraded / Not running / Tampered) with how to fix it. Turning permissions off is detected and the parent is alerted. Uninstalling needs an extra step.	**Alert only:** force stop, turning permissions off, uninstalling, battery-saving and brand-specific app killers. **Not possible:** Safe Mode and "Clear data".

Decision needed – Device Owner mode. If the app is set up during a factory reset (by QR code), most "Alert only" items become **fully prevented**: VPN and Private DNS locked, uninstall blocked, Safe Mode and hotspot disabled. The cost is that the child's phone must be reset. **Our recommendation:** launch as a normal Play Store app and offer Device Owner later as an optional "maximum protection" mode.

POC 1 – Values-based Blur

Area	What the app does	Limitation
Blur inside other apps	Checks the screen about twice a second (YouTube, Instagram, TikTok, browsers). Only the bad picture is blurred, not the whole screen. The blur is real, the page can still be scrolled and tapped, and covers move with the page. Everything happens on the phone: nothing is uploaded or saved.	Android asks for screen-capture permission **again every session**, so it cannot run fully unattended (the parent gets a one-tap restart notice). There is a short delay (under about 1 second) before covering. Apps that block screenshots (banking, Netflix) cannot be checked. It uses a lot of battery.
Parent controls	On/off, categories (Explicit, Suggestive, Gambling, Alcohol & smoking), strictness, blur strength, and mode (bad pictures / whole screen / cover all people).	The current model is good at suggestive pictures and weaker at explicit ones, so a purpose-built model is needed before we claim safety. **Male/female targeting is not reliable**, so we recommend "cover all people".
Browser image search	In Chrome, Edge, Brave and Kiwi, every photo is covered as the page opens and shown only once it is judged clean, using dedicated models for explicit and drug imagery.	Android 14+ only. Explicit and suggestive content only. Photos stay covered in incognito tabs.
Aman's own browser	Images are checked before they are shown, so a bad image never appears. Ads are removed.	Protects only browsing inside Aman's browser.

What we changed after testing: whole-screen covering became per-picture covering; a dark shade became a real blur; flickering covers now move with the page; an 80%-strength cover became full strength using the Accessibility permission.

Permissions Needed

Permission	Used for
VPN (one-time pop-up)	Website blocking, Safe Search, lists, VPN restriction, Internet Pause
Accessibility (Settings)	Search-word blocking, full-strength blur, browser image search
Usage Access (Settings)	Knowing which app is open, closing VPN apps
Display over other apps	Block screen, pause screen, blur covers
Screen capture (every session)	Blur inside other apps
Notifications, Battery, Device admin	Parent alerts, staying active, making uninstall harder

Accessibility, VPN, Usage Access and the full app list are sensitive permissions. Google Play requires a declaration for each, and parental control is an accepted use.